

the fraction of your bankroll that maximizes the growth rate. It is expressed as a formula: $2p - 1 = x$ where 2 times the probability of winning (p) minus 1 equals the percentage of your total bankroll that you should bet (x). For example, if the probability of beating the house is 55 percent, you should bet 10 percent of your bankroll to achieve maximum growth of your winnings. If the probability is 70 percent, bet 40 percent. And if you know the odds of winning are 100 percent, the model would say: Bet it all.

The stock market, of course, is far more complex than the game of blackjack. In a card game, there are a finite number of cards and therefore a limited number of possible outcomes. The stock market, with many thousands of stocks and millions of investors, has an almost unlimited number of various outcomes. Using the Kelly approach requires constant recalculations and adjustments throughout the investment process. Nonetheless, the underlying concept—mathematically linking the degree of probability to investment size—carries important lessons.

I believe the Kelly model is an attractive tool for focus investors. However, it will benefit only those who use it responsibly. There are risks to employing the Kelly approach, and investors would be wise to understand its three limitations:

1. Anyone who intends to invest, using the Kelly model or not, should have a long-term horizon. Even if a blackjack player has a sound model that can beat the house, it does not always hold that success will be revealed in the first few hands dealt. The same is true for investing. How many times have investors seen that they have selected the right company but the market has taken its own sweet time in rewarding the selection?
2. Be wary of using leverage. The danger of borrowing to invest in the stock market (with a margin account) has been trumpeted loudly by both Ben Graham and Warren Buffett. The unexpected call on your capital can occur at the most unfortunate time in the game. If you use the Kelly model in a margin account, a stock market decline can force you to sell your stock, causing you to remove a high-probability bet.
3. The biggest danger in playing high-probability games is the risk of overbetting. If you judge that an event has a 70 percent chance of success when in fact it is only a 55 percent event, you run the risk of